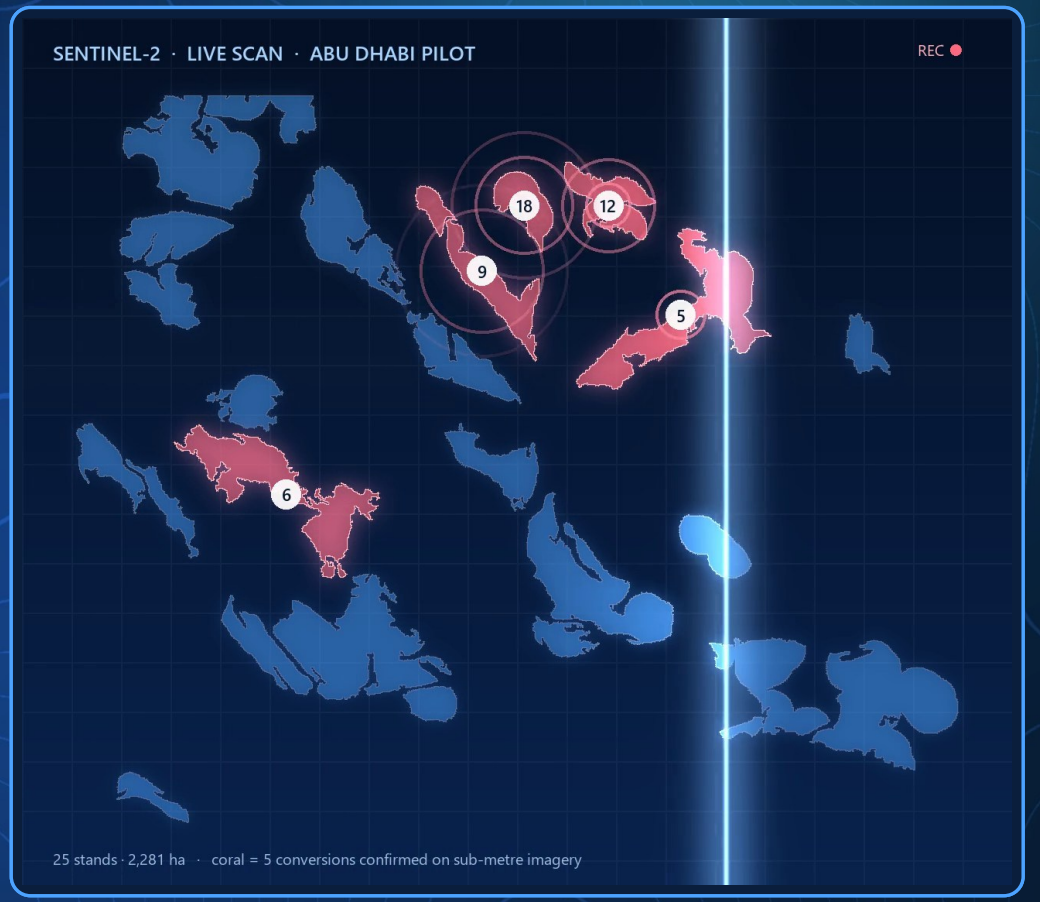
ARAB YOUTH SPACE HACKATHON 2026 · CHALLENGE 813 · FINAL

Blue Carbon Guardian

Tide-aware satellite monitoring for Gulf mangroves — from alert to evidence.

Team Blue Athar · الأثر الأزرق

Theme: Ecosystem Health, Biodiversity & Blue Carbon



91	5	73 %	≈ 1
stands monitored	conversions confirmed	partial loss caught (chance 13 %)	false alarm per stand-year

Nahla Nabil · Bahrain

THE PROBLEM

Restoration is scaling faster than monitoring

100 M — **UAE mangrove pledge by 2030**
COP26, 2021 · Abu Dhabi Mangrove Initiative

100 M+ — **Saudi mangrove pledge by 2030**
Saudi Green Initiative

Feb 2026 — **first GCC mangrove monitoring guide**
EAD + IUCN Mangrove Specialist Group

TODAY field surveys and one-off studies, no standing alert.
WE SOLVE a calibrated alert every ~5 days that says where to inspect and what carbon is at stake.

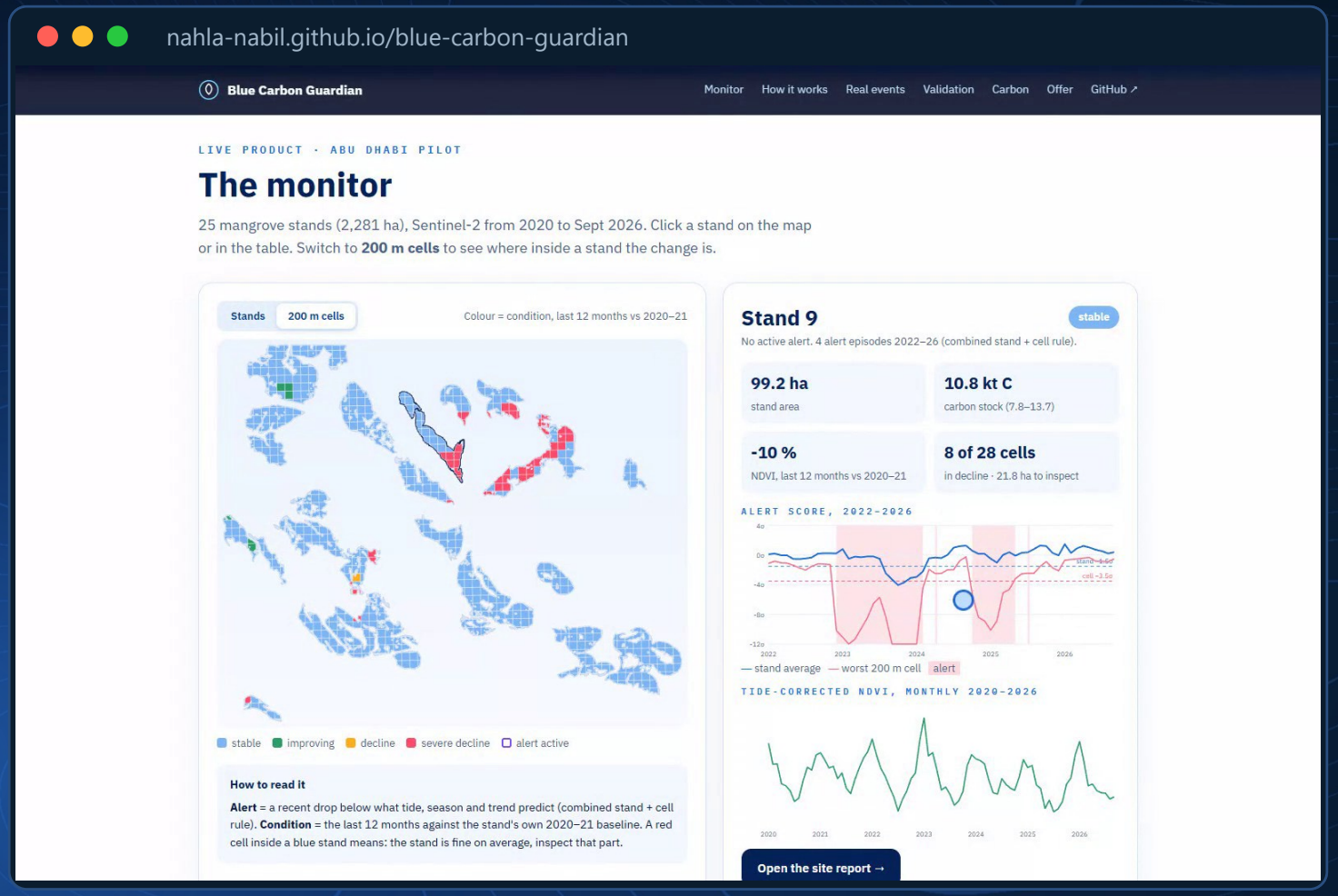
THIS IS HAPPENING NOW · STAND 5, ABU DHABI



Sentinel-2 true colour (contains modified Copernicus Sentinel data). Our alert began on 25 Oct 2022, during the works.

EXAMPLE OUTPUT · THE PRODUCT

From signal to decision, in one screen



One alert
stand OR any 200 m cell, calibrated

Where to look
a 200 m cell map for the inspector

Condition
last 12 months vs own 2020–21 baseline

Carbon at stake
stock with a range, from field data

Site report
one page per stand, every number traced

● LIVE DEMO · open stand 9 on the website

BUSINESS USE CASE & END USERS

Who uses it, and for which decision

Environment agencies

EAD, MOCCAE, Saudi NCVC

THEIR DECISION

Which stands do we inspect this month?

Today: periodic field surveys

monthly alert + 200 m cell map: inspect only flagged cells

Planting & carbon developers

ADNOC, Red Sea Global, carbon projects

THEIR DECISION

Is the carbon we report still there?

Today: plots and drone campaigns

condition + carbon stock with a range, a site report

Developers & regulators

EIA consultants, permit offices

THEIR DECISION

Did works stay outside the mangrove line?

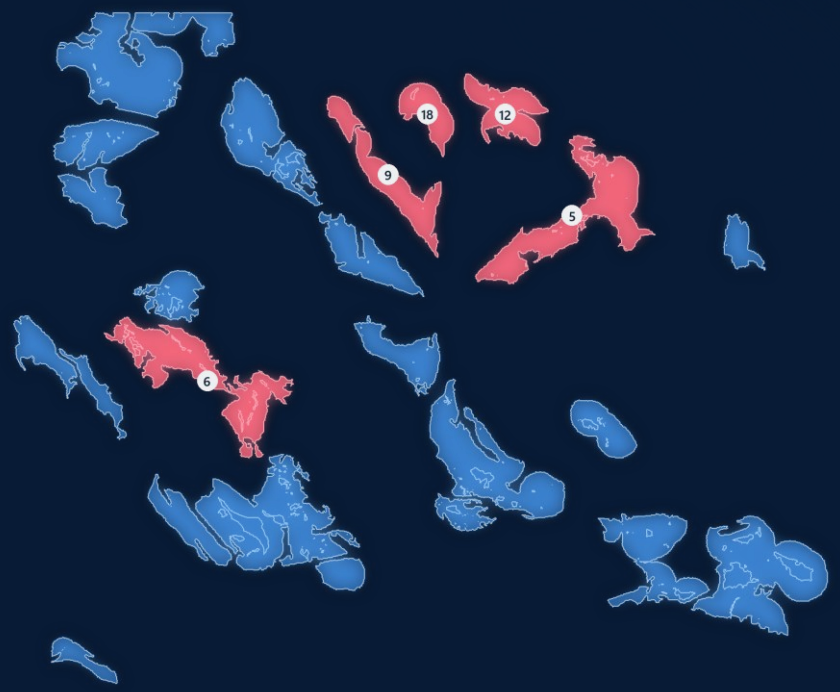
Today: comparing two images by eye

an alert when works enter mangrove cells, dated evidence

New scene every ~5 days → **Alert** ~1 false alarm / stand-yr → **Cell map** where to look → **Inspection** only flagged cells → **Site report** carbon at stake

DATA & STUDY AREA

Five independent data sources, six years, 91 stands



Pilot: 25 stands · 2,281 ha (WorldCover 2021 patches ≥ 3 ha). +66 unseen stands in Abu Dhabi N/W and Saudi Tarut Bay test transfer.

Sentinel-2 L2A
ESA / Copernicus · 10–20 m time series
778 dates

EnMAP hyperspectral
DLR · 224 bands · independent check
2 scenes

ESA WorldCover 2021
10 m land cover · stand outlines
CC BY 4.0

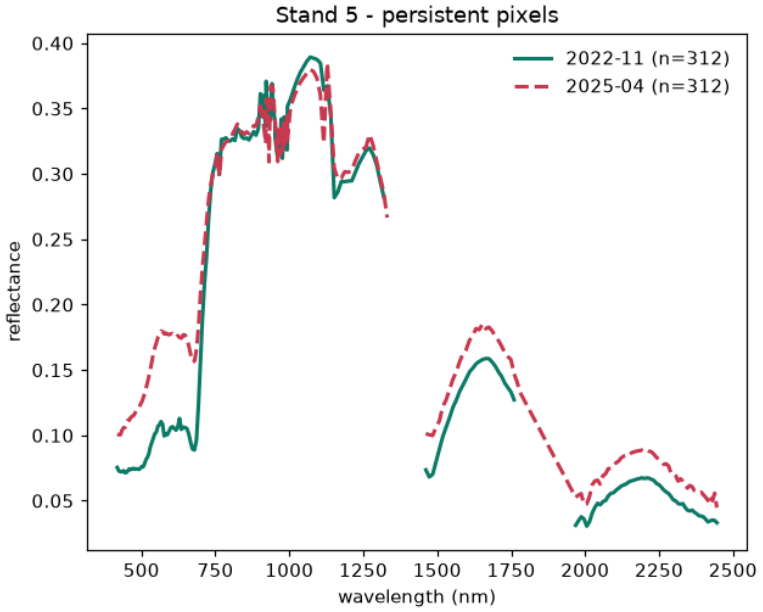
Field carbon plots
Schile et al. 2016 · carbon density
24 plots

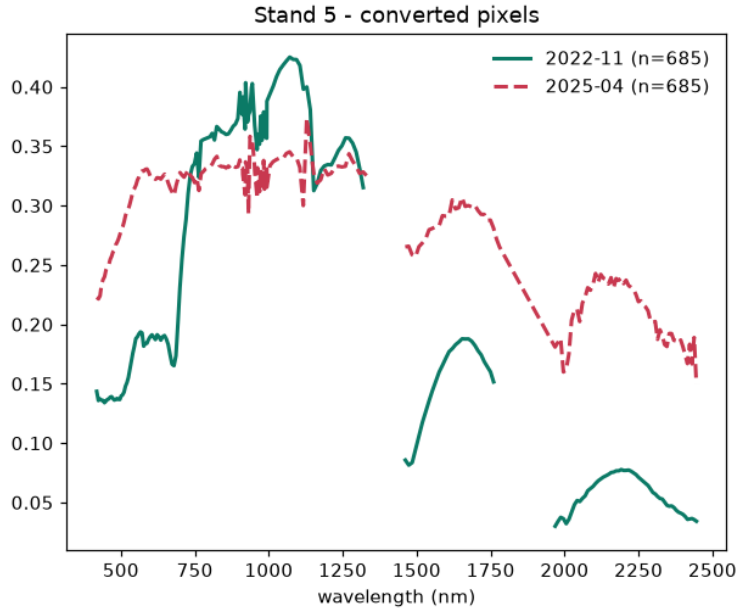
Sub-metre captures
WorldView-2/3, Legion-1 · confirm events
0.3–0.5 m

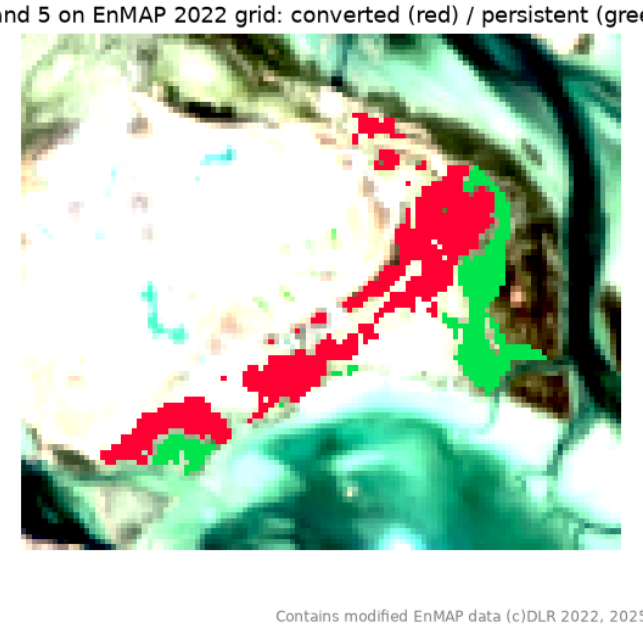
Quality control: cloud masks · interior pixels only · tide model · second-source checks

SPACE DATA · HYPERSPECTRAL (ENMAP, 224 BANDS)

A second sensor shows what the mangrove became







Contains modified EnMAP data (c)DLR 2022, 2025

r = 0.84–0.99

agreement with our Sentinel-2 indices on the same dates (25 and 14 stands)

62 vs 26 ha

stand-5 loss: EnMAP pixel count vs our conservative Sentinel-2 flag

Vegetation → mineral

converted pixels take the spectrum of sand and fill: reclamation, not dieback

Honest limit: as a classifier, the full spectrum is no better than Sentinel-2-like bands (F1 0.74 vs 0.72). Contains modified EnMAP data © DLR 2022, 2025.

APPROACH · HOW IT WORKS

From satellite pixels to an inspection decision

Stands & 200 m cells

WorldCover 2021 ≥ 3 ha · interior pixels only

Sentinel-2 series

778 dates · cloud mask · NDVI, red edge, moisture, wetness

Tide-aware model

trend + season + tide terms, refitted every month

Alert & condition

stand ≤ −1.5σ OR any cell ≤ −3.5σ · last 12 months vs 2020–21

Decision layer

cell map · carbon at stake · dashboard · site reports

+ Field carbon · 108 t C/ha

+ Checks · EnMAP · sub-metre

HOW A USER RECEIVES IT

New scene · every ~5 days → Alert?

yes → Check flagged cells on sub-metre imagery → Email / JSON API: cell map + carbon at stake → Inspector visits only the flagged cells

no → Monthly site report: condition, trend and carbon stock for every stand

TECHNICAL INNOVATION 1

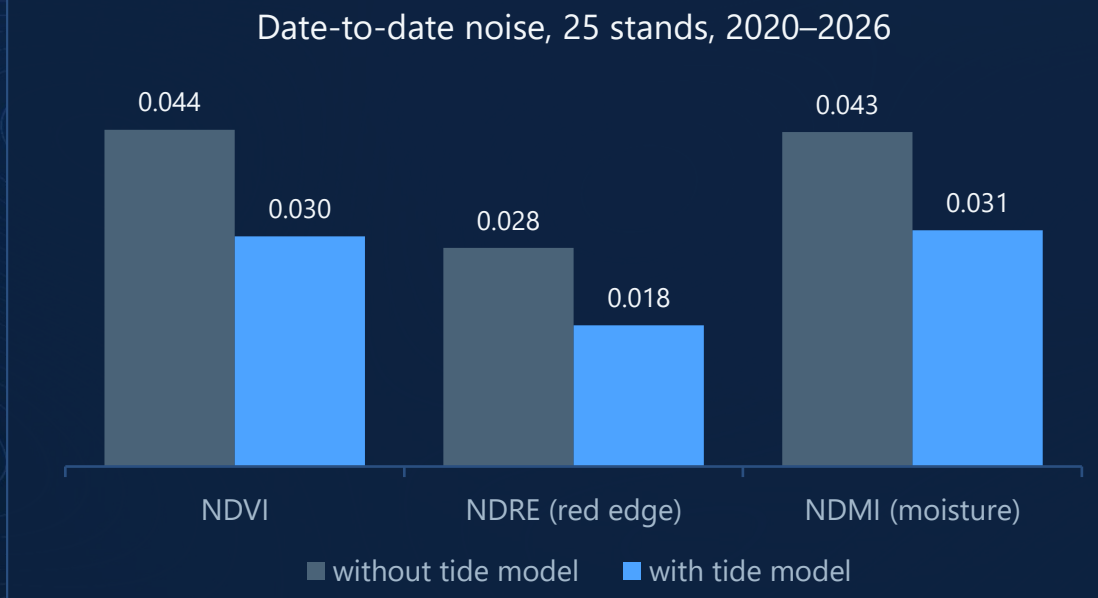
The tide looks like tree loss — until you correct for it

WHY NAIVE MONITORS FAIL ON TIDAL FLATS



NDVI **0.30** NDVI **0.18**

Same trees, same season. Measured on our stands: noise vs wetness r ≈ −0.9.



−29 to −35 %

We regress every index on trend, season and tidal state before flagging anything. Noise falls on every index.

TECHNICAL INNOVATION 2

A stand average hides partial loss. 200 m cells do not.



Real stand-5 data, 2022 → 2026: cells turn red while the stand average barely moves.

Partial loss caught within 60 days (simulated)

73 % stand + cells

25 % stand alone

13 % by chance

8 of 28

cells of stand 9 in severe decline while its average reads "stable"

3 of 5

real conversions found only by the cells (stands 12, 18, 6)

ONE ALERT RULE

stand ≤ −1.5σ **OR** any cell ≤ −3.5σ

calibrated together: 0.96 false-alarm episodes per stand-year

How we differ from what exists today

	Field surveys	One-off satellite maps	Free loss alerts (Global Mangrove Watch)	Blue Carbon Guardian
How often	campaigns	yearly or less	monthly	**every ~5 days**
Corrects for the tide	n/a	not usually	not documented	**✓ tide model**
Says where inside a stand	✓	—	—	**✓ 200 m cells**
Published false-alarm rate	n/a	n/a	accuracy only	**✓ ≈ 1 per stand-year**
Carbon at stake per alert	—	—	—	**✓ field-based**
Gulf coverage	✓	✓	not confirmed	**✓ Abu Dhabi + Tarut Bay**

Complementary, not competing: Global Mangrove Watch flags loss worldwide; we add the tide model, the 200 m cells, a measured false-alarm rate and the carbon at stake — built and tested for the Gulf.

VALIDATION 1 · REAL EVENTS

Five real conversions, confirmed on sub-metre imagery



Stand 5 reclamation with finger canals



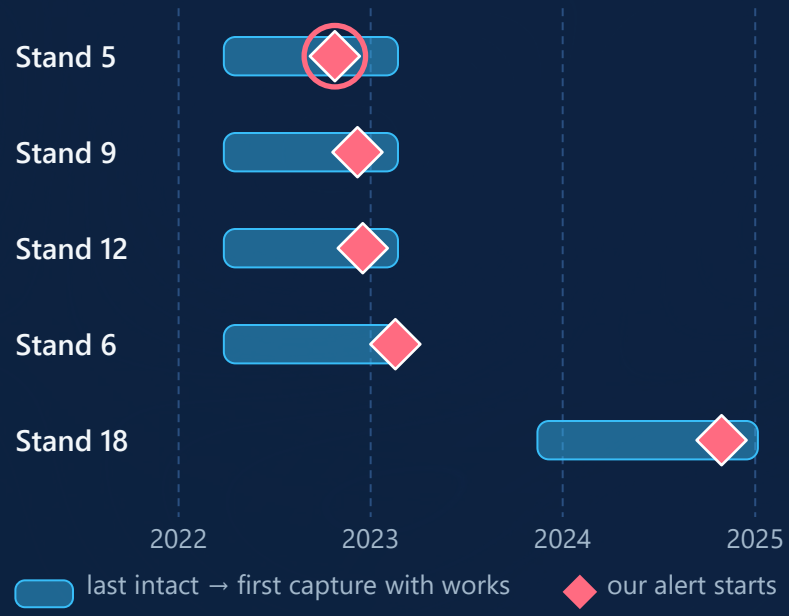
Stand 9 canal development, south-east



Stand 12 found only by the cells



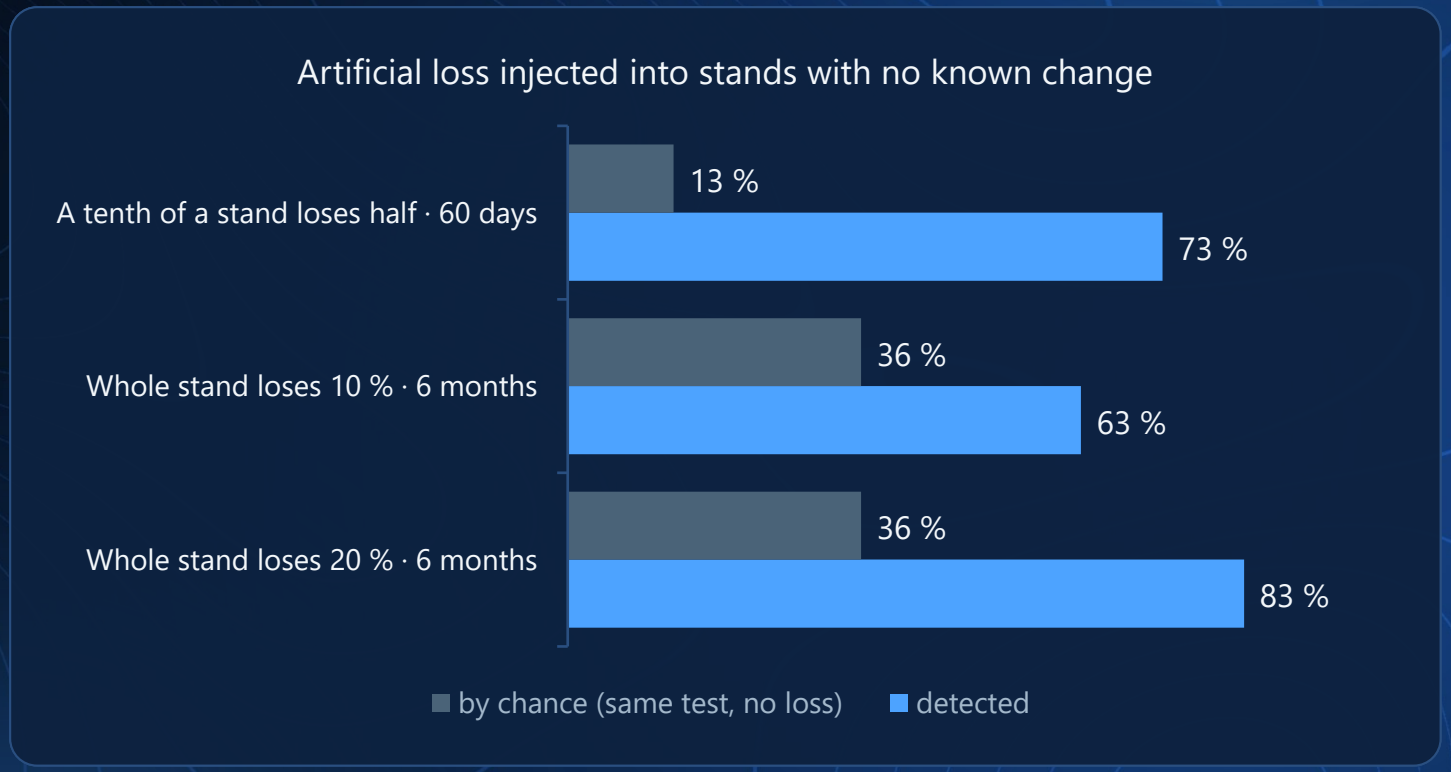
Stand 18 found only by the cells, 2025



Every alert began inside the window. Detection as works happen, not advance warning.

VALIDATION 2 · CHANCE, UNSEEN STANDS, SECOND SENSOR

Every rate next to its chance rate



A detection counts only if a new alert starts after the loss begins. ~160–240 trials per row.

CALIBRATED · 25 STANDS

0.96

false-alarm episodes per stand-year

UNSEEN · 65 STANDS

0.79

per stand-year, 305 stand-years, incl. Saudi Tarut Bay

ENMAP · SAME DATES

r = 0.84–0.99

vs our Sentinel-2 indices; stand-5 loss 62 ha vs our 26 ha

What we do not claim: advance warning, field verification, species discrimination, or that hyperspectral classifies better (F1 0.74 vs 0.72). We withdrew our own first test (58 % / 90 %): it had no chance baseline.

IMPACT & BUSINESS VIABILITY

Carbon at stake, and who pays to protect it

108 t C/ha

field density near the stands (90 % CI 79–138); 82 % of it is in the soil

247 kt C

pilot stock, 2,281 ha (range 180–316) ≈ 907 kt CO_2e

2.8–6.7 kt C

stock on the 26–62 ha lost at stand 5: what one alert puts on the table

SERVICE TIERS (PRICES ARE PILOT HYPOTHESES)

Monitor

$12k

per site / yr

alert, cell map, condition, carbon, monthly reports

Evidence

$25k

per site / yr

+ sub-metre check of every alert, audit pack

Development watch

$8k

per project / yr

for coastal developers and EIA consultants

Cost to serve ≈ $3.3k / site / yr: compute < $5 (measured), sub-metre checks $500–750, analyst $2k (assumption).

COMPETITION

Global Mangrove Watch sends free monthly loss alerts (Gulf coverage not confirmed). We are complementary: tide-aware, calibrated, stand + cell, carbon and reports.

THE HONEST MARKET

Gulf + Red Sea subscriptions ≈ $0.2–0.4 M / yr. The real prize: a **national MRV contract** or a monitoring layer on **gIQ**. SDG 13 · 14 · 15.

BEYOND THE PROOF OF CONCEPT

Reproducible today, built to scale tomorrow

DOCUMENTATION & REPRODUCIBILITY

Public repository, MIT licence
code, notebook, figures, provenance and licences

Re-runs end to end in ~24 s
outputs committed · pinned requirements · example in/out

Clean-machine test passed
fresh environment, same numbers, 0 errors

Every number traceable
each figure comes from a script and a written summary

Live website + site reports
anyone can open the monitor and check a stand

SCALABILITY

Proven now

91 stands · 2 countries · 2 Sentinel-2 tiles · 65 unseen stands at 0.79 false alarms / stand-yr · compute < $5 per site per year

Incubation

Satellite 813 in the spectral step · deployment on gIQ · a pilot customer + site records · more EnMAP dates

Scale

Gulf ≈ 165 km^2 + Red Sea ≈ 200 km^2 of mangrove · sub-metre data for small stands (Bahrain) · national MRV, licensing

Areas: Frontiers in Marine Science 2025. Compute measured; prices assumed.

THE TEAM

One engineer, every role — delivered in the open

Nahla Nabil

Team lead · Bahrain

RELEVANT EXPERTISE

- AI / LLM backend engineering
- Machine learning & computer vision (deepfake detection)
- Federated learning · semantic search systems
- Python geospatial stack

COMMUNITY

NASA Space Apps Challenge local lead: organising and mentoring space-data teams

HOW ONE PERSON DELIVERED THIS

An AI-assisted engineering workflow (Claude Code) that I direct and review; every number is reproducible from the public code.

ROLES & RESPONSIBILITIES IN THIS PROJECT

Project lead & owner

problem framing, user needs, scope, every claim

EO data engineer

778 Sentinel-2 dates, EnMAP reader, sub-metre checks

Modelling & statistics

tide-aware model, stand + cell alert, calibration

Validation & integrity

real events, simulations, unseen stands

Product & delivery

dashboard, website, site reports, notebook

Business & impact

customers, pricing hypotheses, carbon ranges

Incubation gap we will close: a mangrove ecologist mentor · a restoration or agency partner · a GIS analyst

TEAM BLUE ATHAR · الأثر الأزرق

Every lost mangrove leaves a trace. We find it from space.

A working product · five real events confirmed · every number reproducible

Tide-aware	5 of 5	0.79	Open
29–35 % less noise	conversions alerted during the works, 3 only by cells	false alarms / stand-yr on unseen stands	public code, re-runs in ~24 s

Our ask: incubation to run a pilot with a Gulf agency or planting programme, mentor access on hyperspectral reading of mangroves, and Satellite 813 scenes when available.

nahla-nabil.github.io/blue-carbon-guardian · github.com/Nahla-Nabil/blue-carbon-guardian

Thank you

The 2-minute product demo

